

Principles of multifunctional agriculture for supporting agriculture in metropolitan peri-urban areas: The case of Greater Melbourne, Australia

Ana Spataru^a, Robert Faggian^a, Annemaree Docking

^a Centre for Regional and Rural Futures, CeRRF, Deakin University, 75 Pigdons Road, Waurin Ponds, 3216, Victoria, Australia

ARTICLE INFO

Keywords:

Multifunctional agriculture
Peri-urban agriculture
Multifunctionality principles
Greater Melbourne

ABSTRACT

Agriculture in metropolitan peri-urban areas is influenced by structural changes associated with increased input costs, an aging farming population, difficulty in retaining labour, and fluctuations in market conditions. In the absence of clear planning policies, the proximity to the city has fuelled land fragmentation, encroachment, and increased land value, thus increasing pressure on existing farms. Using the case of Greater Melbourne in Victoria, we argue that metropolitan planning policies have diminished the agricultural potential of peri-urban agriculture by creating a precedent for more urban uses in otherwise agricultural landscapes. The future of peri-urban agriculture requires alternative models that address emerging issues of resilience and viability. The concept of Multifunctional Agriculture (MFA) presents a useful model to re-evaluate metropolitan peri-urban agriculture beyond food production functions by integrating the environmental and socio-cultural role into local and regional economies. Based on the adopted definition and broad characterisation of MFA as implemented in the European Union, we discuss a set of six principles of multifunctionality: *flexibility*, *collaboration*, *smaller-can-be-better*, *long-term strategies*, *use of technology*, and *circular resource use*. These principles aim to expose the fundamental processes required for transforming agriculture in Australian metropolitan peri-urban areas.

1. Introduction

Metropolitan peri-urban areas (MPUa) are rural spaces where structural changes such as price fluctuations, succession, aging farmers and productivity are amplified by proximity to urban centres (Butt, 2013; Houston, 2005). From the rural perspective, MPUa are the *invaded countryside* (Bourne et al., 2003), while from the urban perspective, they represent a stock of land and resources waiting to satisfy urban needs (Friedberger, 2000). This has been referred to as the *double vulnerability* of peri-urban agriculture (Rojo et al., 2014).

In MPUa across all Australian cities, agriculture is a marginal economic activity contributing less than 1% to local economies (Rawnsley, 2017). Discussions about how MPUa should support future agricultural development are hampered by the dichotomy between urban and rural land use, lack of agreement over the role of these spaces, oversimplification of existing relationships, and conflicts and inadequate institutional and organisational structures for agricultural development (laquinta and Drescher, 2000). However, peri-urban agriculture offers multiple economic and non-economic benefits. Economic benefits include local employment, local agribusiness growth, multiplier effect and access to a variety of markets without higher transport costs (Carey et al., 2016). Non-economic benefits include ecosystem functions

(water, soil and air quality; and biodiversity), recreation and heritage, community coherence, quality of life, education and health opportunities, and tourism (Brinkley, 2012; Brown and Reeder, 2007; Hellerstein et al., 2002).

Multiple policies and tools are used to protect farmland from urban expansion and encourage farming activities in MPUa. Broadly, they include zoning, agricultural buffers, 'Right to farm' ordinances, acquisition of farmland through trusts, financial incentives, taxation, and national legal regulations (Richardson, 2007; Sokolow, 2004). They are differently applied in the European Union (Perrin et al., 2018; Romero and Melo, 2015), the United States (Brinkley, 2012) or Canada (Bryant and Chahine, 2016). In Australia, given the long history of agricultural reform, mainly zoning and market-based tools are used (Bunker and Houston, 2003; Carey et al., 2011; Harman et al., 2015). Importantly, the actors involved in the process of decision-making play a vital role in how MPUa farmland is valued. Invested economic interests and discursive power can influence the narrative of farmland protection and can remove some stakeholders from shaping the discourse of peri-urban farmland valuation (Vinge, 2018).

The economic and planning model in Australian MPUa favours the paradigm of continuous urban growth and is ill-equipped to capture the economic and non-economic benefits of agriculture (McFarland, 2015).

* Corresponding author.

E-mail address: aspataru@deakin.edu.au (A. Spataru).

<https://doi.org/10.1016/j.jrurstud.2019.11.009>

Received 10 April 2019; Received in revised form 23 October 2019; Accepted 5 November 2019

Available online 20 November 2019

0743-0167/ © 2019 Elsevier Ltd. All rights reserved.

Issues around climate change, sustainable food production, food security, landscape preservation and resource management are not captured by current business-as-usual models. Alternative models are necessary to capture how farmland will be valued in the coming decades in the wake of future drivers of structural changes – technology, consumer demand, policy, international trade, environmental quality – and climate change (ABARES, 2007; Buxton et al., 2011).

Victorian State and Local Governments have committed to promote sustainable agriculture through multiple strategies, policies, and programs (e.g. Smarter, Safer Farms; Young Farmers; Food Innovation Network; Green Wedge Management Plan; Planning for sustainable animal industries; <http://agriculture.vic.gov.au>). However, the sustainable agriculture framework is not in itself sufficient to ensure landscape transformation. Multifunctional agriculture (MFA) represents an agricultural model that re-evaluates agricultural activity and is prescriptive in nature. Through the MFA lens, a holistic planning framework can be developed that includes food security, environmental, economic, and socio-cultural functions (FAO, 1999; Renting et al., 2005). In Australia, a few attempts to define MFA conceptually have been noted but it remains an elusive concept in practice (Dibden and Cocklin, 2007; Holmes, 2006).

In essence, MFA is a *post-productivist* agricultural model moving beyond *competitive productivism* characterised by monocultures, industrial-scale production, and expansion of world food trade and supply chains (Halfacree, 1997; Mathera et al., 2006). Some have argued that MFA is disguised protectionism. This is owing to the European Union adopting the MFA model and advocating for special treatment at World Trade Organisation negotiations, arguing that MFA supports European environmental sustainability and rural development (Potter and Burney, 2002). Regardless, agriculture in MPUa seems to be confronted by similar challenges irrespective of country, making MFA a suitable model for rural transformation (Lovell, 2010; Piore et al., 2018; Simon, 2008). There is no singular definition or approach to MFA due to contextual competing environmental, economic, and socio-cultural objectives; for example, the commodification of rural landscapes through tourism (Tonts and Grieve, 2002; Wu, 2018). However, MFA is aligned with sustainable development and community resilience through synergies between economic objectives and environmental and socio-cultural practices.

To explore the applicability of MFA in Australian MPUa, we are using the case of Greater Melbourne to, first, discuss the influence of past metropolitan planning policies (MPP) on peri-urban agriculture, and second, propose a set of *multifunctionality principles* able to inform an alternative trajectory for peri-urban agriculture in Australia.

2. The Australian agricultural sector

2.1. Brief overview

Over 85% of the Australian population lives within 50 km of the coast (ABS, 2004). This influences farm size, with smaller farms in MPUa and farms of thousands of hectares in regional Australia. Agricultural land use patterns are determined by climate variability, water availability, soil type, and proximity to markets (Malcolm, 1996). According to the Australian Bureau of Statistics (ABS), in 2017/18 the total gross value of agricultural production in Australia was \$58.9 billion with approximately 85,500 operating farms (on a declining trajectory from 145,000 in the last 20 years). The state of Victoria contributed a quarter of the total gross value of agricultural production, with \$14.9 billion from 21,000 operating farms (ABS, 2017c). Australia is a net exporter of primary products, with livestock and meat being the fastest growing segment (79% increase in the last 20 years). Australian farmers consider the expected population growth and rising income in Asia a great emerging business opportunity (Jackson et al., 2018).

The Australian agricultural sector promotes the neoliberal ideology that farmers need to manage risk, uncertainty, and markets on their

own rather than being supported through governmental subsidies (Lockie, 2009). This ideology is the result of a long history of agricultural reforms between 1970 and 2000 (Gray et al., 2014). The main messages were ‘increase production volume’ and ‘get big or get out’, making *competitive productivism* the norm. As a result, in the last 40 years, farms with revenues greater than \$1 million have increased from 3 to 16% and occupy 63% of farmland, from 45% (Jackson et al., 2018).

Broadly considering farming practices across Australia, there is evidence of concern over the future of natural environments and farmer's livelihoods under the *competitive productivism* paradigm (Pritchard and McManus, 2000). To become more efficient, farmers have extended the use of fertilisers, insecticides, veterinary pharmaceuticals, and fossil fuel dependent farm equipment (Argent, 2002; Heasman and Lang, 2015). The use of irrigation and clearing of natural vegetation has allowed farming to expand in areas less suitable or less accessible for agriculture (Cullen, 2005; Lawrence et al., 2013). Intensification, specialisation, and economic concentration allowed Australian farmers to increase productivity and efficiency, often at the expense of the environment (Argent, 2002; Dibden et al., 2009). From an environmental perspective, some argue that the government has delivered short-term solutions to mounting evidence of ill-suited agricultural practices that resulted in soil erosion, soil salinization, pest plants and animals intrusion, and damage to native vegetation (Gray and Lawrence, 2001). *Competitive productivism* creates dependency on commercial seed and agro-chemical inputs, displaces traditional knowledge and practices, and can favour economic and development arguments that discourage practices that support ecosystem services, mitigate the impacts of climate change and increase agricultural resilience (Kimbrell, 2002; Vandermeer et al., 2002).

2.2. Agriculture in the metropolitan peri-urban areas – Greater Melbourne

MPUa are unique spaces where urban and rural features create rapid changes in land use patterns (Scheromm and Soulard, 2018). The notion of peri-urban itself is case specific and ‘cannot be easily defined or delimited through unambiguous criteria’ (OECD, 1979). These spaces attract debate over future urban growth and agricultural activity, as well as the identity of rural spaces in an increasingly urbanised and globalised society (Cadieux and Hurley, 2010; McCarthy, 2008). The processes that have affected agriculture in MPUa of Greater Melbourne are loss, dilution, transition, and transference (Buxton et al., 2007).

- Loss – diminishing farmland as a result of conversion to urban and commercial use. The converted land is costly and impractical to revert back to agricultural use;
- Dilution – unclear delimitation of urban and rural land uses creates demand for rural lifestyle living. Farmers are squeezed out due to encroachment, fragmentation, increased land value, and incompatibilities between commercial farming and residential land use;
- Transition – fewer businesses in the agricultural sector result in a diminishing cluster effect, particularly evident in the horticultural and dairy industries. Diminished ‘critical mass’ further reduces the support of upstream industries (input providers, machinery services) and downstream industries (manufacturing, distribution);
- Transference – relocation to areas outside the peri-urban as a consequence of limited expansion possibilities or financial viability.

In short, compared with Regional Victoria, agricultural production in Melbourne's MPUa has been relatively stable. Agricultural activity is characterised by high-value production, fewer large scale agribusinesses, and a decrease in the number of operating agribusinesses (Butt, 2013).

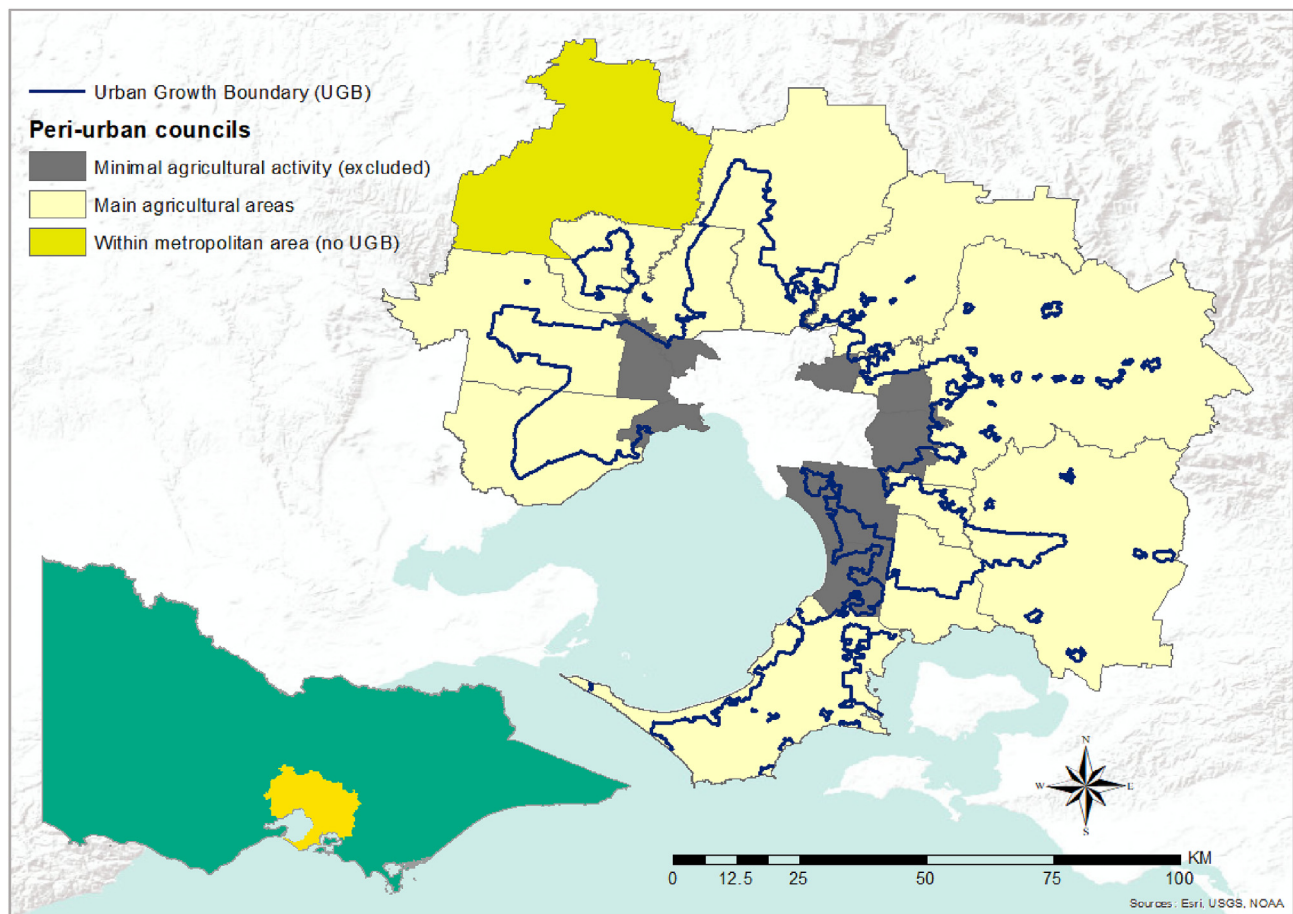


Fig. 1. The metropolitan peri-urban areas of Greater Melbourne (Source: Australian Bureau of Statistics – Statistical geography).

Table 1

Areas identified by MMBW with considerable value for agriculture, conservation, recreation, and resource management (Source: MMBW, 1971):

Area considered	Functions considered
1 – Yarra Valley	Water pollution control, management and recreation, scenic landscape, and agriculture
2 – Dandenong Ranges	Intensive agriculture and scenic landscape
3 – Dingley Area	Important for coarse sand deposits
4 – Melbourne Airport	Transport
5 – Maribyrnong Valley	Significant flora and fauna, recreation potential, and market gardens
6 – Derrimut	Open rural lands and stock holding saleyards and abattoirs
7 – Point Cook, Mornington Peninsula	Recreation purposes along the shoreline

3. The influence of metropolitan planning policies on agriculture

In this section, we will analyse in more detail the impact of past and current MPP on agriculture in MPUa. We argue that policy and market forces have put peri-urban agriculture on a trajectory incompatible with emerging challenges. In this analysis, we consider as MPUa those metropolitan Local Governments currently divided by the urban growth boundary (UGB) but within the metropolitan boundary of Greater Melbourne (Fig. 1), also referred to as the inner peri-urban areas. An existing outer peri-urban area extends further beyond the metropolitan area (Carey et al., 2016), however it is not the focus of this paper. We suggest focusing on this immediate MPUa because the pressures of land fragmentation, land banking, and land speculation are much stronger and potentially create a precedent for conversion of farmland further

into the outer peri-urban areas.

3.1. The history of metropolitan Melbourne planning policies

The urban form of Greater Melbourne has a radial shape, following the major transport routes with designated growth corridors and open green spaces separating urban areas. The purpose of open spaces is to ensure that non-urban land use – agriculture, conservation, landscape, and recreational open space – will be maintained around Greater Melbourne. The first and most comprehensive MPP was published by Melbourne and Metropolitan Board of Works (MMBW) in 1971, ‘Planning Policies for Melbourne Metropolitan Region’. The Report was followed by the ‘Metropolitan Strategy Implementation’ in 1981, which provided the policy context of urban consolidation (Buxton and Tieman, 2004).

The Report identified seven areas with important non-urban functions (Table 1), which later became Greater Melbourne’s 12 green wedge zones across 17 Local Governments. The Report proposed the introduction of planning zones to enforce protection from urban development, designating five zones: Conservation Zone, Landscape Interest Zone, Special Extractive Zone, Intensive Agriculture Zone, and General Farming (MMBW, 1971).

The Strategy further acknowledged that the metropolitan countryside is a region of strategic importance for the liveability of urban residents and the prosperity of the city:

These wedges ensure that scenic landscapes, farming areas, native vegetation and wildlife habitats are preserved within the metropolitan area. These areas help satisfy people’s needs for fresh vegetables and other primary products and provide opportunities for hobby farming and areas

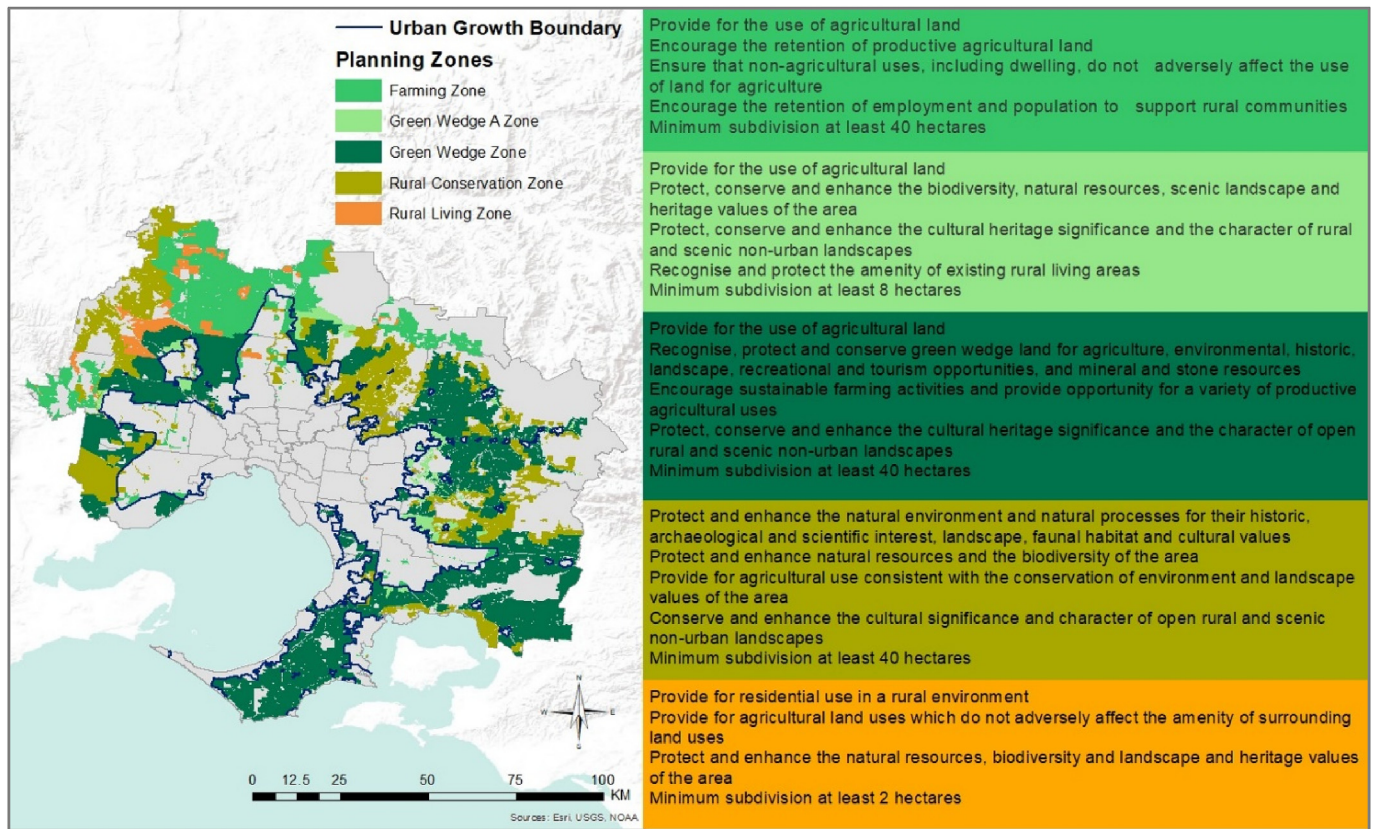


Fig. 2. Metropolitan planning zones related to agriculture and rural living (Source: Victoria Planning Provisions).

where people can ‘escape’ from the city. (Metropolitan Strategy Implementation, pp.85, 1981)

According to the Strategy, the intention was to ‘eliminate urban expectation from rural areas, to encourage rural enterprises and to facilitate the use of particular areas in accord with their land capability’ (MMBW, 1981, p. 85), with objectives to:

- (1) Implement planning provisions for rural areas designed to dissuade subdivision and to offer a thriving environment for rural enterprises;
- (2) Define minimum allotment size to acknowledge that sound agricultural activities needed appropriate scale;
- (3) Match land capability with rural land use through planning zones in a way that minimises environmental degradation (water pollution, soil erosion, and air pollution).

The subsequent MPP, ‘Shaping Melbourne’s Future’ (published in 1987) further supported the vision set by MMBW, emphasising key valuable areas needed appropriate consideration.

The retention of wedges of countryside between the urban corridors ensures that scenic landscapes, farming areas, native vegetation and wildlife habitats are preserved within the metropolitan area. The ‘green wedges’ provide Melbourne with primary produce, and opportunities for rural living and recreation. Existing governmental policies protect environmentally sensitive areas in the Upper Yarra Valley, the Dandenong Ranges, The Mornington Peninsula and the Macedon Ranges from excessive urban growth. (Shaping Melbourne’s Future, pp.4, 1987)

Despite 16 years since the initial Report by MMBW and further commitment to protect farmland in 1987, urban consolidation fuelled land speculation, pollution, and outward growth. Greater Melbourne was expanding via a south-eastern axis where key valuable non-urban land had previously been identified. In an attempt to control this

inevitable force, the following MPP, ‘Living suburbs’, published in 1992, focused solely on channelling urban growth towards the north-west and managing congestion, pollution, and housing affordability. Agricultural aspects remained completely unaddressed, setting a precedent for a narrow focus on urban issues without consideration of wider impacts on other types of land use (Tsutsumi and Wyatt, 2006). Shortly after, in 1996, the Victoria Planning Provisions (VPPs) were introduced. The previous five non-urban zones were replaced with three rural zones – Rural Zone, Environmental Zone, and Rural Living Zone. The VPPs also included less prohibited land uses, which meant that a wider range of urban-related uses could be considered by Local Governments. Planning permits facilitated evaluation of case-by-case applications, which resulted in ‘quasi-urban’ uses in the green wedges, including industrial, commercial, retail, and major tourism developments (Buxton and Goodman, 2002).

‘Melbourne 2030: Planning for sustainable growth’, published in 2002, reignited concern over the future role of green wedges. The most important outcomes through *Direction 2* were the introduction of the UGB and the Green Wedge Management Plan (GWMP) guidelines to be used by the 17 Local Governments. It was recognised that past policies have done little to curb loss of MPUs farmland and more attention should be paid to managing rural living:

Issues related to the management and planning of green wedges will have to be re-examined. This will require the development of revised criteria for rural living developments, and their incorporation into local planning decisions [...] Management plans will need to be developed to address the unique development issues affecting each green wedge. (Melbourne 2030: Planning for sustainable growth, pp.173, 2002)

The latest MPP, ‘Plan Melbourne: 2017–2050’ was published in 2017 and marks a significant departure in scope from previous MPPs. Firstly, the Plan focuses on a polycentric urban form, with city centres linked via ‘a network of clusters, centres, precincts and gateways’ (Principle 3)

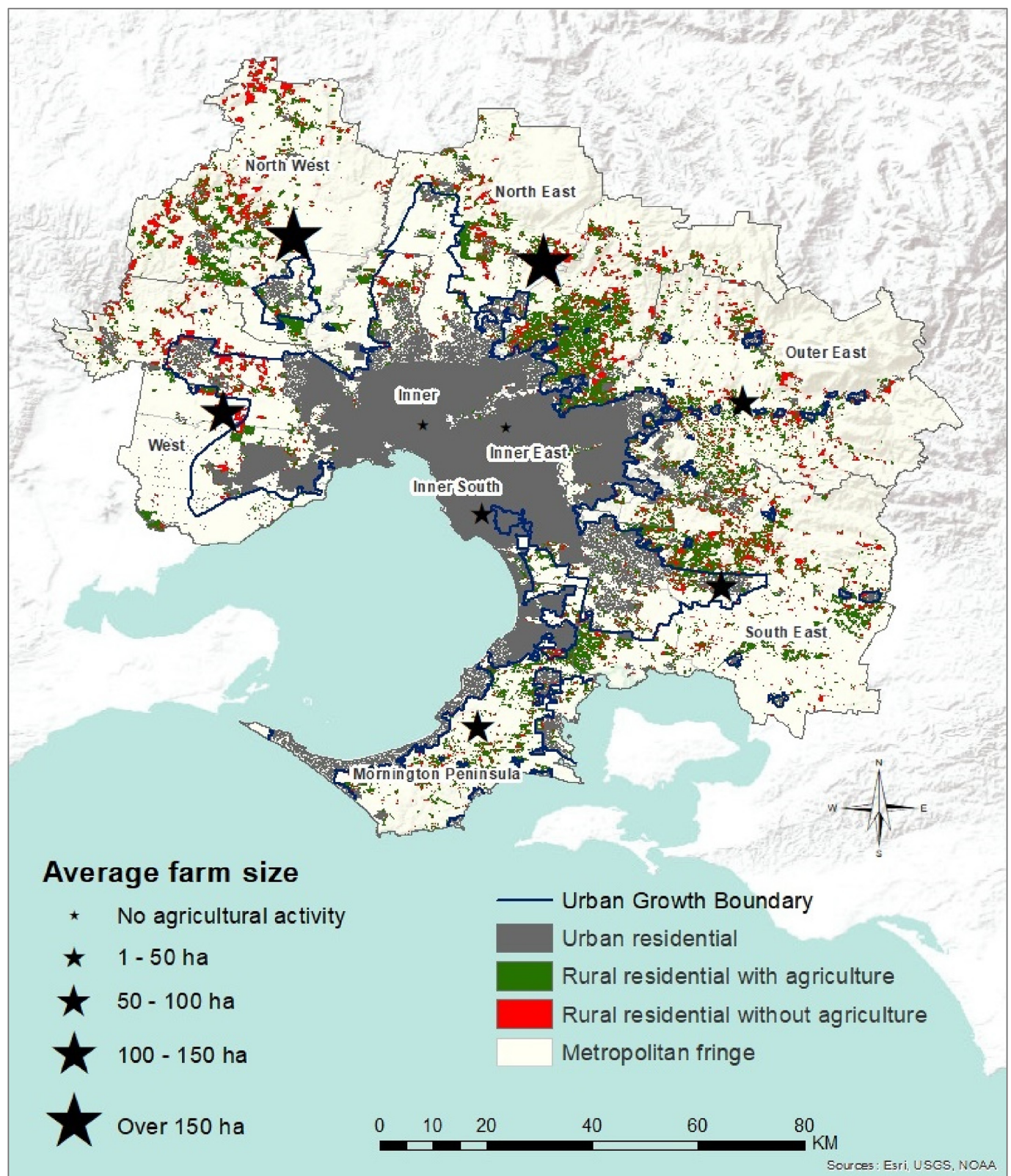


Fig. 3. Average farm size (physical component) and the distribution of urban-rural residential land use at the metropolitan level (Source: Australian Land Use and Management, 2017, Australian Bureau of Statistics –Agricultural Commodities Census, 2017–18).

and on urban planning designed around the 20-min neighbourhoods concept (Principle 4). Secondly, the peri-urban areas of Greater Melbourne are specifically mentioned alongside green wedges, thus suggesting recognition of their character (*Direction 4.5: Plan for*

Melbourne's green wedges and peri-urban areas). Lastly, protection of key agricultural areas has a wider scope than economic concerns and includes the preservation of landscape value and biodiversity (DELWP, 2017).

Currently, there are four planning zones related to agricultural activity and one rural living. Fig. 2 summarises the corresponding planning zones across MPUa and their purpose – latest update (VPP, 2018).

The expansion of Greater Melbourne via a south-eastern axis is due to more pleasing landscapes for urban development and higher agricultural suitability in terms of soil type, water availability, climate and environmental value for farming activities, especially high-value horticulture. The average farm size in the south-east is half the size of farms in the north-west (Fig. 3). Between 2007/08 and 2017/18, land fragmentation, urban development pressure, land banking, and structural changes resulted in higher loss of farmland in Greater Melbourne (23% - from 252,000 ha to 194,000 ha) compared with Regional Victoria (13% - from 12.3 million ha to 10.7 million ha) (ABS, 2017b). For the same period, the value of agricultural production in Greater Melbourne increased by 24% from \$1.22 billion to 1.51 billion, while in Regional Victoria it increased by 28% from \$10.44 billion to \$13.4 billion (ABS, 2017c).

3.2. The impact of metropolitan planning policies on the farming community

There is evidence that economic growth and socio-economic aspects of MPUa play an important role in how land, and ultimately agriculture, are valued (Jaquinta and Drescher, 2000). Despite restrictive land use in MPUa, large insurance, finance and development companies are undermining the planning process through large land purchases, opening an avenue towards land banking (Senate, 2016). Also, increased interest in rural living has fuelled land speculation, described by Sandercock (1990, p. 187) as the ‘national hobby’. Land speculation is detrimental to agriculture since farmers can become property developers themselves that regard their land as a superannuation asset and are less likely to invest in long-term agricultural management. As the agricultural capability of land diminishes, it becomes an excuse for urban development, which creates a cycle of speculation and diminished agricultural investment (Buxton and Goodman, 2002). While the MPP since 1954 have set ambitious goals and actions to protect agriculture in MPUa, there are significant contradictions within them. For example, designated growth areas render commitments to limit urban expansion irrelevant and new rural zones promote land banking for urban, commercial, and industrial uses. Coupled with the VPP's inefficient, complex, vague, and bulky framework, the negative impacts on agriculture continue to mount (Buxton and Goodman, 2014).

Based on farmers' testimonies as part of the Parliamentary Inquiry (PI) into the Sustainable Development of Agribusiness in Outer Suburban Melbourne in 2009, it is evident that efforts to contain urban expansion did not result in protection of farmland. Moreover, the MPPs described above only created an uncertain and discontent farming sector. According to some farmers:

Green Wedge planning provisions are negative. With the district drifting between declining broadacre land use and no clear direction for future agricultural development, existing land owners are disadvantaged by Green Wedge provisions. Can't sub-divide and exit and can't attract buyers for large properties as land prices are forced up by demand for lifestyle farms. [...] Without any active support, vision for the future and resources from all levels of government, I am struggling to find positive arguments for investing in agriculture in the district. (David Nickell, Bruce Ure, Neil Firrito, from Gembrook: PI, 2009)

Other farmers have concerns over the future of horticultural potential due to diminished critical mass, lack of horticulture infrastructure, conflicts with neighbours, pest infestation, change in microclimates due to heat absorbing urban materials, inadequate irrigation and water supplies, and high production costs concluding that:

The principle land use in the non-urban zone is residential not agricultural with little industrial or extractive use. To a large degree the “country” in Council's motto [Manningham] is about landscape and

environmental management and not agricultural management. That distinction needs to be honestly recognised in the planning controls and creatively addressed as opposed to leaving landowners, like us, carrying the burden of landscape managers (Aumann Family, Warrandyte: PI, 2009)

The MPPs are filled with discourses about food security, jobs and development of the agricultural sector, however, there is unclear direction for supporting farmers in MPUa. Land use amendments evaluated on an individual basis allowed for the progressive build-up of urban development pressure leaving decisions about the future urban growth and land use at the hands of landowners, developers, and Local Governments (Buxton and Goodman, 2002).

As urban development gradually encroaches, the viability of farming these fringe areas will continue to decline [...] Land values will continue to rise, making the possibility of land amalgamation impossible to achieve. Attempting to lock in agricultural production in these areas will only condemn us to gradual decline over time through operational inefficiency, with many land holders only holding on in the hope that one day there will be a change of policy and the land rezoned for urban development. (J&JM Schreurs and Sons, Clyde: PI, 2009)

Not surprisingly, the number of agribusinesses has decreased since 2009, by 52%, from 3024 to 1995. Much of this decline occurred for farms with an annual turnover size of less than \$50,000 (ABS, 2017b). In one recent case study of MPUa land suitability, Spataru et al. (2018) highlighted that farmers have concerns over their farm viability and land stewardship. They raised concerns over lack of irrigation water, climate variability, underdeveloped markets for novel products, encroachment, increased local government rates, and lack of agricultural scale.

Farming in the peri-urban is important to the future sustainability of the region. [...] It needs to be acknowledged that a farm must be economically sustainable or viable, and if a business struggles with its economic viability, eventually there is an impact. In the case of farming, the impact may be on the maintenance of the property, the environmental stewardship or the quality of the product (Judy Clements, Victorian Farmers Federation, Whittlesea Branch, 2017)

Many of the issues raised by farmers require systematic intervention rather than piecemeal regulation. The role of farming in MPUa can be re-evaluated through alternative models of agriculture. New functions of peri-urban agriculture have been described by Buxton et al. (2007) and vary from purely commercial production (intensification, aggregation) to environmental, recreational and aesthetic integration (hybridisation, gentrified). In line with these functions, MFA represents an agricultural model aligned with new societal demand for environmental protection, food security, preservation of cultural traditions, and promotion of rural development (Van Huylenbroeck et al., 2007; Wilson, 2007).

4. Multifunctional agriculture for effective transformation of agriculture in MPUa

The usefulness of principles for discussing agricultural development lies in the attractiveness that they are discoverable rather than invented. In this way, principles reproduce the type of content considered ‘true’ in a significant way. Principles also represent generalities, as opposed to particularities, thus making them practical to conceptualise agricultural development in line with multifunctional agriculture (Pretty, 2008).

4.1. Characterisation of multifunctional agriculture

According to the Food and Agriculture Organisation (FAO) the functions addressed by MFA include food security, environmental,

economic, and socio-cultural functions. Environmental functions consider positive and negative externalities from agricultural production through provision of public goods such as biodiversity, landscape, water management, and rural amenities, focus on conservation practices, and seek to reduced chemical inputs (Maier and Shobayashi, 2001). Economic functions capture primary production of products and services with wider economic effect for the farm and the local community, including goods for the food market, foods of distinctive quality and services such as tourism, social care, education, and energy. Socio-cultural functions account for people's livelihoods, cultural values, and the viability of rural communities ranging from social cohesion, employment, identity, heritage and settlement patterns to ethical functions such as animal welfare and fair trade (FAO, 1999; Renting et al., 2009). Broadly, MFA promotes an agricultural model that can address both farm viability and landscape transformation (Marsden, 2003; Wilson, 2007).

Conceptualising MFA in the wake of multiple applications, scales and approaches remains difficult since the inter-relationships between economic, environmental, social, cultural, and institutional dimensions are still being understood, and are often case specific (Renting et al., 2009). The Organisation for Economic Co-operation and Development (OECD) assigns two interpretations to MFA (Maier and Shobayashi, 2001). One interpretation is the *positive concept of multifunctionality*, where the agricultural outputs or effects are multiple and interconnected ('jointness' of production). Some outputs are valued in existing markets, while others may represent externalities not accounted for by the market. This is a purely economic interpretation concerned with emerging relationships among producer, consumer, and taxpayer. The other interpretation is the *normative concept of multifunctionality*, where agriculture is entrusted with fulfilling certain additional functions in society. From this societal interpretation, MFA's focus shifts towards setting objectives that consider issues of equity, stability, viability, and livelihood. We argue that these two interpretations are not mutually exclusive; rather, they both contribute to explain the range of implications for policy-makers, farm development, farmer's identity, academic research, and rural development.

van der Ploeg et al. (2002) have proposed farm expansions along three dimensions favouring the *normative concept of multifunctionality*:

- (4) Broadening – expansion across multiple markets, existing or emerging. Strategies available to farmers include provision of environmental services through various arrangements and contracts (e.g. land conservation, sale of energy), recreational on-farm activities (e.g. accommodation, catering, hunting, trekking, animal riding), educational activities (e.g. workshops for students, craft activities), care activities (e.g. farm therapy for disadvantaged groups of people);
- (5) Regrounding – integration of farms within a wider economic framework with the purpose of providing alternative income arrangements. Strategies include off-farm employment and energy production for farm use;
- (6) Deepening – incorporation of non-conventional farming activities and innovations in product and process development. Strategies available to farmers include development of products with unique attributes (e.g. organic, niche products, high quality products), on-farm processing, direct selling and marketing (e.g. local markets, pick-your-own, e-commerce), and more ethical farming practices.

The usefulness of broadening dimension in MPUa should be considered in the wider applicability of metropolitan land use planning for effective integration of agricultural activities with urban land uses. Many authors, including (Renting et al., 2005; Sturzak and Mell, 2017) argue for increasing metropolitan plurality of land uses and identities to create a *multifunctional landscape* and reduce the "hygienic mode of regulation" characteristic of land use planning (Horlings and Marsden, 2011).

The adoption of MFA requires that all levels of government recognise the role played by MFA for sustainability, farm viability, landscape preservation, and land stewardship. In Norway, France, and Italy, this is achieved by allowing for the legitimacy of new contract arrangements, private-public arrangements, and regional jurisdiction flexibility (Rønningen et al., 2018; Van Huylenbroeck and Durand, 2003). Governance influences the ability of farmers to capitalise on emerging markets through diversification of farm activities and possible synergies between MFA functions. Not all farms have the same opportunities for transitioning to MFA (Wilson, 2008). Case studies in The Netherlands and Italy highlighted that successful adoption of MFA also depends on the type of economic activity and size, farmer's age and education, geographical characteristics, and perceived roles in the rural community (Casini et al., 2012; Meraner et al., 2015; Pfeifer et al., 2009). Farmers are entrepreneurs rather than producers, constantly identifying current and emerging opportunities based on strategic decisions that respond to increased complexity of the business, market fluctuations, and the impacts of climate change (Methorst et al., 2017; Morgan et al., 2010). However, Burton and Wilson (2006) have shown that breaking out from the *competitive productivist* thinking is extremely challenging. Farmer's self-definition continues to be deeply rooted in the *productivist* mode despite farming activities that would correspond to a multifunctional model of agriculture.

In the UK, USA, and Australia, conflicting paradigms shaping the MFA discourse of agricultural restructuring through de-regulation in capitalistic governance in general, and European rural development in particular, continue to challenge wide implementation of MFA (Marsden and Sonnino, 2008; Tilzey, 2006). One paradigm is based on narrow economic reasoning addressing the issue of scale and specialisation of production. MFA translates into a survival strategy, while maintaining agro-industrial practices and the technological treadmill, particularly small-scale and less productive farms (Marsden and Sonnino, 2008). Another paradigm is based on arguments of environmental protection and amenity preservation focused on landscape transformation through farm diversification into ecological, social, and aesthetic functions that contribute to the farmer's income (Fish et al., 2008). The last paradigm is based on sustainable development objectives where agricultural activity is an agent of change for how agri-food systems, rural economies, and rural cultures are organised (Marsden and Sonnino, 2008). This paradigm is difficult to translate into policy since governments and agencies for rural development are limited by jurisdiction, capabilities, resources, and diverging interests among different agricultural stakeholders. Institutions are also not designed to manage integrated complexity, departments often working in isolation as Clark (2006) evaluated in England. Thus, limitations for landscape transformation through MFA are governmental and institutional in nature. At the farm level, limitations include perceptions around freedom and independence, trust in government, economic gain from direct sales and on-farm processing, consumer demand, market access, and diversification opportunities (Jongeneel et al., 2008). The dichotomy between individual freedoms influenced by incentives and trade-off and the moral imperative for ecosystem protection and preservation yields different types of evidence of how MFA can and should shape rural development in the future (Hodge, 2004, p. 342).

4.2. Principles of multifunctional agriculture

The integrative nature of MFA results in a variety of definitions as shown by Van Huylenbroeck et al. (2007). However, based on the characterisation presented above, a broad definition of MFA is required before presenting the principles of multifunctionality. We define MFA as:

An encompassing planning framework and practice based on the production of food (commodities) and services (non-commodities) that contribute to the socio-economic viability of rural spaces while practicing

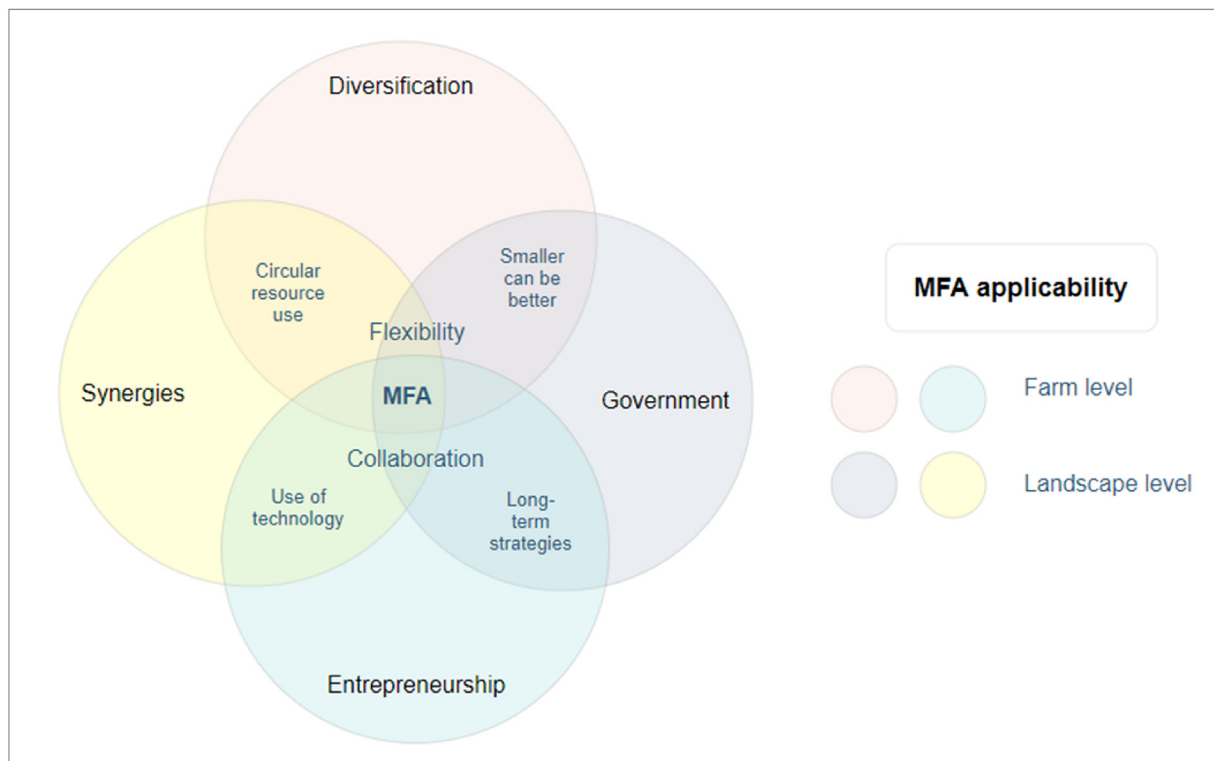


Fig. 4. Inter-relationships between MFA characteristics and principles (author's diagram).

sound ecological and ethical management with positive impacts on employment, biodiversity, food safety and security, cultural and historical heritage, and rural landscape.

The principles proposed are cross-dimensional and seek to simplify the complexity that exists among the MFA functions. We also draw on examples where MFA expression is occurring in Greater Melbourne's MPUa. We define six principles: *flexibility*, *collaboration*, *smaller can be better*, *long-term strategies*, *use of technology*, and *circular resource use* (Fig. 4). These principles are influenced by the applicability scale. Diversification and collaboration address farm level transformation, while governance and synergies promote landscape transformation.

The flexibility principle refers to the ability of farmers, government agencies, and associations to change contractual obligations according to external influences based on principles of fairness (Rønningen et al., 2018). Foreseen and unforeseen changes act as driving forces for innovation, transfer of knowledge, and partnerships (Seuneke et al., 2013). This is only possible when agriculture is not narrowly defined by commodity production but includes societal and environmental goods and services. This increases the complexity of the farm business strategy and MFA promotes dynamic agricultural systems able to respond to emerging risks and opportunities in a way that contributes to local economic development (Hendrickson et al., 2008a, Hendrickson et al., 2008). The *double vulnerability* of agriculture is already the driver of multiple farm adjustments in Greater Melbourne, including land set aside for conservation, water, and biodiversity (Landcare Program), on-farm processing, accommodation and tourism activities (mainly in areas of wine production), and niche production. However, many of these activities occur either in isolation or where capacity already exists such as in Yarra Valley and Mornington Peninsula. To encourage uptake of MFA at the landscape level, governmental agencies and various associations in the health, education, recreation or management sectors could implement new processes to reduce bureaucracy and re-evaluate contracts on a yearly (or period specific) basis. In part, existing pressures from high land valuation could be alleviated by adoption of more flexible arrangements between small-scale farms. Examples include the

Hedge Row Farm in north-west Greater Melbourne and state-wide pilot program Farm Co-operatives and Collaboration (Farming Together), which aims to foster local food supply chains.

The collaboration principle highlights that agriculture does not operate in a vacuum. Multifunctionality is based on creating commodity and non-commodity goods and services that have a positive outcome for farm viability and rural development. However, stepping outside the family farm can be extremely challenging (Seuneke et al., 2013). Thus, collaboration between multiple stakeholders helps build relationships, share knowledge, discuss common issues, and discover new business opportunities and strategies. Based on the history of MPPs in Greater Melbourne's MPUa presented so far, it is clear that collaboration between Local Government and farmers needs to improve. Support for other food related industries, particularly food processing, is also required, as The Food and Beverage Growth Plan for Melbourne's North indicates a struggling food processing industry (McKinna and Wall, 2014). Efforts to situate agriculture in a wider context that addresses issues about health, resilience, environment, and food security have been undertaken by the Victorian Health Promotion Foundation with the purpose of creating a Food Policy Coalition, however it is not tailored to MPUa in particular (Loff et al., 2010). In subtler ways, there is a level of disconnection found in the form of fewer children learning first-hand about agriculture and food production or having fewer relatives in the farming sector. These children become tomorrow's decision makers unaware of the realities faced by many farmers in MPUa. Creating the right collaboration space is, in theory, easier in Greater Melbourne than Regional Victoria since expertise in business, law, research, policy, and alternative agricultural practices is more extensive. Collaboration also builds a common vision, direction for the future, and narratives that can shift economic policy towards more integrative evaluation of the role of peri-urban agriculture.

The smaller-can-be-better principle refers to moving agricultural activities from economies of scale to economies of scope and synergy. Small-scale farmers in Australia have proven to be versatile, innovative and adaptable (Hollier and Reid, 2007). In a market-driven

environment where ‘get big or get out’ defines the farmer’s main business strategy, alternative models are proving equally relevant, including MFA, sustainable intensification, regenerative agriculture, and climate-smart agriculture (FAO, 2013; Petersen and Snapp, 2015; Wilson, 2007). Agriculture in Greater Melbourne’s MPUa operates on an average farm size of 100 ha compared with 550 ha in Regional Victoria, while overall agricultural output is \$7,000 per hectare compared with \$1,000 per hectare in Regional Victoria (ABS, 2017c). Small-scale farms in MPUa have more opportunities to focus on non-commodity goods and services such as education, recreation, ecosystem conservation, and eco-tourism due to proximity to the city, thus creating a new identity that values the multi-faceted roles of farming in MPUa (Holmes, 2008).

The *long-term strategies* principle takes under consideration that environmental constraints, climate change, and rural development requires vision and commitment from all agricultural stakeholders. Government policies and supporting programs can vary depending on political shifts, however, to realise MFA, strong leadership and commitment is needed (Clark, 2006). The history of MPPs tells a story of decision making that over time has diminished the function, role, and purpose of peri-urban agriculture. The initial MPP in 1971 recognised the need to apply a uniform planning strategy regardless of the inherent differences between green wedges across Greater Melbourne. However, this sound strategy has been weakened by assessing each green wedge area on specific primary purposes, thus favouring protection of some areas over others and creating uncertainty for those farmers in less valued areas (Buxton and Goodman, 2002). Moreover, instability at the fringe due to urban growth pressures is reducing investment in long-term projects with potential of transforming the rural landscape. As a result, farmers are mainly reactive rather than proactive as indicated by the farmer’s statements. Farming is, in essence, a risky commercial activity dependent on multiple external factors outside the control of the farmer. Long-term strategies result in greater resilience, knowledge, and preparedness. For example, Agribusiness Officers in Mornington Peninsula and City of Whittlesea have uncovered that agricultural opportunities are underestimated and their officer role is to provide essential guidance for long-term development strategies (Flett, 2016).

The *use of technology* principle refers to opportunities available to farmers to address economic and environmental challenges more efficiently. Sassenrath et al. (2008) argues that new technology should be adopted according to farmer’s needs and avoid creating a digital technological treadmill. Agricultural technological advancement is broad, varied and rapid. It ranges from simple improvements such as electrified fencing and modern irrigation systems to the use of Information and Communication Technology (ICT). ICT, through management and decision making technologies, coupled with satellite and high resolution data collection and analysis, has the potential to inform important business strategies. Off-farm employment is also made possible through remote working arrangements and high speed internet, allowing for income diversification without leaving the farm. To this end, Agriculture Victoria is testing a \$12 million On-Farm Internet of Things, mainly engaging with regional farms, providing the network coverage and test devices that monitor soil moisture, chemical application, tank levels, and livestock health. Dufty and Jackson (2018) showed that farmers already use ICT for internet commerce (40%) and to a lesser extent for online presence through social media (5%). In MPUa there are less limitations facing the uptake of ICT compared with regional areas (Paül and McKenzie, 2013). However, issues of cost, training, learning and sharing opportunities, and scale of operation, results in the lowest percentage of ICT compared with other business sectors nationwide (ABS, 2017d).

The *circular resource use* principle refers to farming systems that reduce waste and emissions, building on circular economy objectives to maximise use and reuse of resources (the technical cycle), while minimising degradation and depreciation of natural resources (the biological cycle) (Breure et al., 2018; EMF, 2017). Realising environmental constraints and operating based on sound ecological principles must

guide most farming activities (Hendrickson et al., 2008a, Hendrickson et al., 2008b). Closing waste cycles offers the benefits of reducing input costs, assisting natural nutrient cycles, and minimising dependency on agro-chemical products (Jurgilevich et al., 2016). In this context, MFA offers the route to sustainability and encourages solutions that adapt the farm to the environmental conditions rather than adapting the environment to consumerism. Multiple opportunities exist and are already being applied by farmers in MPUa. Such opportunities include organic waste that can be reintegrated as compost or soil enhancer, use of renewable energy that will reduce the overall dependency on fossil fuels, and wastewater that can be reused as recycled water for irrigation purposes. Sound ecological practices have clear economic benefits ranging from access to niche markets (organic, high quality, locally sourced) to reduced production costs (Pagotto and Halog, 2016). In MPUa, farmers adopted many farming practices on motives of caring for the natural environment. For example, mixed pasture cultivations are widespread (35,000 ha), trash management includes practices such as crop stubble retained on ground or standing (46%), incorporated in soil (13%), and grazed off (13%), all considered more sustainable. The use of compost, mulch and poultry manure accounts for 25% of soil enhancer use (data for Port Philip and Western Port Natural Resource Management Region (ABS, 2017a)). The most promising resource reuse is recycled wastewater from urban residential sources. Currently, in the west of Melbourne, recycled water accounts for 51% of all agricultural water use, and 17–18% in south-east and Mornington Peninsula (ABS, 2019). These areas have long been identified for their agricultural potential, highlighting that when agriculture is supported it becomes a space for innovation and investment. Population growth in Greater Melbourne is expected to reach 8 million by 2051, with dwellings increasing to 3.2 million by 2051 (DELWP, 2016). Assuming that future population growth is channelled in designated growth areas, the additional recycled wastewater can support peri-urban agriculture in areas currently farming without access to irrigation water, for example in the north, where on-farm dams are the main source of irrigation water.

Expressions of multifunctionality at the farm level are more easily identified, whereas coherent policies across different scales have proven difficult to implement (Clark, 2006; Hassink et al., 2016). The design of policies remains the most useful avenue to promote MFA and the principles presented in this paper aim to offer some support. Despite the MFA model being developed and applied in the European Union context, it can benefit the Australian MPUa facing population growth, concerns over food production capacity, security, and safety, and environmental degradation. Arguably, some limitations include the political, policy, and discourse environment and the apparent lack of incentives for transformational action. However, the strengths of the MFA model are its adaptability to a wide range of agricultural systems and landscapes.

5. Conclusion

In this paper we presented an overview of the vulnerability of agriculture in metropolitan peri-urban areas using the case study of Greater Melbourne. Peri-urbanisation in Greater Melbourne has occurred at the expense of open space, productive agricultural land, environmental functions, and farm viability. MPPs have recognised the need to protect agriculture and the environment as early as the post-war era. Efforts to channel urban expansion within growth corridors separated by green wedges started in 1971, however, much of the ambitious objectives of past policies have been diminished through development pressures, incremental land use amendments, and lack of commitment from Local Governments. This has created a precedent for valuing agriculture only in economic terms, neglecting many other roles of agricultural landscapes.

The farming community in MPUa of Greater Melbourne have proven particularly resilient to political and market conditions. The value of

production and agricultural output continued to increase. However, the agricultural landscape is highly fragmented. For the continuation of farming in MPUa, the existing farming community needs to be encouraged and supported. Multifunctional agriculture represents an agricultural model in line with sustainability. The principles of multifunctionality: *flexibility, collaboration, smaller can be better, long-term strategies, use of technology, and circular resource use*, offer an insight into what is needed for transformative action in MPUa by focusing on opportunities and existing innovations.

Declaration of interest statement

No potential conflict of interest of our paper.

Acknowledgements

We express our appreciation to the anonymous reviewers and colleagues that suggested improving this paper over time.

References

- ABARES, 2007. Drivers of Structural Change in Australian Agriculture. Rural Industries Research and Development Corporation, Canberra.
- ABS, 2004. 1301.0 - Year Book Australia 2004. Australian Bureau of Statistics [accessed January, 2019].
- ABS, 2017. 4627.0 - Land Management and Farming in Australia 2016-17. Australian Bureau of Statistics [accessed November, 2018].
- ABS, 2017. 7121.0 - Agricultural Commodities 2015-16. Australian Bureau of Statistics [accessed November, 2018].
- ABS, 2017. 7503.0 - Value of Agricultural Commodities Produced 2016-17. Australian Bureau of Statistics [accessed November, 2018].
- ABS, 2017. 8129.0 - Business Use of Information Technology, 2015–16. Australian Bureau of Statistics [accessed July, 2019].
- ABS, 2019. 4618.0 - Water Use on Australian Farms, 2017-18. Australian Bureau of Statistics [accessed July, 2019].
- Argent, N., 2002. From pillar to post? In search of the post-productivist countryside in Australia. *Aust. Geogr.* 33 (1), 97–114.
- Bourne, L.S., Bunce, M., Taylor, L., Luka, N., Maurer, J., 2003. Contested ground: the dynamics of peri-urban growth in the Toronto region. *Can. J. Reg. Sci.* 26 (2 and 3), 251–270.
- Breure, A.M., Lijzen, J.P.A., Maring, L., 2018. Soil and land management in a circular economy. *Sci. Total Environ.* 624, 1125–1130. <https://doi.org/10.1016/j.scitotenv.2017.12.137>.
- Brinkley, C., 2012. Evaluating the benefits of peri-urban agriculture. *J. Plan. Lit.* 27 (3), 259–269. <https://doi.org/10.1177/0885412211435172>.
- Brown, D.M., Reeder, R.J., 2007. Farm-based recreation: a statistical profile, Economic Research Report Number 53, United States Department of Agriculture. Economic Research Service. https://www.ers.usda.gov/webdocs/publications/45941/12871_err53.pdf?v=0.
- Bryant, C.R., Chahine, G., 2016. Action research and reducing the vulnerability of peri-urban agriculture: a case study from the Montreal Region. *Geogr. Res.* 54 (2), 165–175. <https://doi.org/10.1111/1745-5871.12119>.
- Bunker, R., Houston, P., 2003. Prospects for the rural-urban fringe in Australia: observations from a brief history of the landscapes around Sydney and Adelaide. *Aust. Geogr. Stud.* 41 (3), 303–323. <https://doi.org/10.1046/j.1467-8470.2003.00236.x>.
- Burton, R.J.F., Wilson, G.A., 2006. Injecting social psychology theory into conceptualisations of agricultural agency: towards a post-productivist farmer self-identity? *J. Rural Stud.* 22 (1), 95–115. <https://doi.org/10.1016/j.jrurstud.2005.07.004>.
- Butt, A., 2013. Exploring peri-urbanisation and agricultural systems in the Melbourne region. *Geogr. Res.* 51 (2), 204–218. <https://doi.org/10.1111/1745-5871.12005>.
- Buxton, M., Butt, A., Farrell, S., Alvarez, A., 2011. Future of the fringe: scenarios for Melbourne's peri-urban growth. In: *The State of Australian Cities Conference*, Melbourne.
- Buxton, M., Goodman, R., 2002. Maintaining Melbourne's Green Wedges: Planning Policy and the Future of Melbourne's Green Belt. School of Social Science and Planning. RMIT University.
- Buxton, M., Goodman, R., 2014. The impact of planning 'reform' on the Victorian land use planning system. *Aust. Plan.* 51 (2), 132–140. <https://doi.org/10.1080/07293682.2014.892866>.
- Buxton, M., Tieman, G., 2004. Urban Consolidation in Melbourne 1988-2003. RMIT Publishing, Melbourne.
- Buxton, M., Tieman, G., Bekessy, S., Budge, T., Butt, A., et al., 2007. Change and Continuity in Peri-Urban Australia. Peri-Urban Case Study: Bendigo Corridor, Canberra (Land and Water Australia).
- Cadieux, K.V., Hurley, P.T., 2010. Amenity migration, exurbia, and emerging rural landscapes: global natural amenity as place and as process. *Geojournal* 76, 297–302.
- Carey, R., Krumholz, F., Duignan, K., McConell, K., Browne, J., et al., 2011. Integrating agriculture and food policy to achieve sustainable peri-urban fruit and vegetable production in Victoria, Australia. *J. Agric. Food Syst. Commun. Dev.* 1 (3), 181–195. <https://doi.org/10.5304/jafscd.2011.013.003>.
- Carey, R., Larsen, K., Sheridan, J., Candy, S., 2016. Melbourne's Food Future: Planning a Resilient City Food Bowl. The University of Melbourne: Victorian Eco-Innovation Lab.
- Casini, L., Contini, C., Romano, C., 2012. Paths to developing multifunctional agriculture: insights for rural development policies. *Int. J. Agric. Resour. Gov. Ecol.* 9 (3/4), 185–203. <https://doi.org/10.1504/IJARGE.2012.050347>.
- Clark, J., 2006. The institutional limits to multifunctional agriculture: subnational governance and regional systems of innovations. *Environ. Plan. C Govern. Policy* 24, 331–349. <https://doi.org/10.1068/c053>.
- Cullen, P., 2005. Water: the key to sustainability in a dry land. In: Goldie, J., Douglas, B., Furnass, B. (Eds.), *In Search of Sustainability*. CSIRO Publishing.
- DELWP, 2016. Victoria in the Future. Population and Household Projections to 2051. Victoria State Government Retrieved from. https://www.planning.vic.gov.au/_data/assets/pdf_file/0030/97608/Victoria-in-Future-2016-FINAL-web.pdf.
- DELWP, 2017. Plan Melbourne 2017-2050. Victoria State Government: Department of Environment, Land, Water and Planning.
- Dibden, J., Cocklin, C., 2007. Contesting competition: governance and farmer resistance in Australia. In: Cheshire, L., V H, Lawrence, G. (Eds.), *Rural Governance: International Perspectives*. Routledge, Abingdon, pp. 175–190.
- Dibden, J., Potter, C., Cocklin, C., 2009. Contesting the neoliberal project for agriculture: productivist and multifunctional trajectories in the European Union and Australia. *J. Rural Stud.* 25 (3), 299–308. <https://doi.org/10.1016/j.jrurstud.2008.12.003>.
- Duffy, N., Jackson, T., 2018. Information and Communication Technology Use in Australian Agriculture: A Survey of Broadacre, Dairy and Vegetable Farms. Department of Agriculture and Water Resources: Australian Bureau of Agricultural and Resource Economics and Sciences.
- EMF, 2017. Concept: what Is a Circular Economy? A Framework for an Economy that Is Restorative and Regenerative by Design. [accessed July, 2019].
- FAO, 1999. Cultivating our futures: the multifunctional character of agriculture and land. In: Paper Presented at the FAO/Netherlands Conference, Maastricht, The Netherlands, Abstract retrieved from. <http://www.fao.org/docrep/x2777e/x2777e00.htm#TopOfPage>.
- FAO, 2013. Climate-Smart Agriculture Sourcebook. Food and Agriculture Organisation of the United Nations.
- Fish, R., Seymour, S., Steven, M., 2008. Sustainable Farmland Management: Transdisciplinary Approaches. CABI, Wallingford, UK; Cambridge, MA.
- Flett, S.J., 2016. Agribusiness Extension Officers/Food Hubs Review. Sustain The Australian Food Network, Melbourne.
- Friedberger, M., 2000. The rural-urban fringe in the late twentieth century. *Agric. Hist.* 72 (2), 502–514.
- Gray, E.M., Oss-Emer, M., Sheng, Y., 2014. Australian Agricultural Productivity Growth: Past Reforms and Future Opportunities. ABARES, Canberra.
- Gray, I., Lawrence, G., 2001. A Future for Regional Australia: Escaping Global Misfortune. Cambridge University Press, Cambridge.
- Halfacree, K., 1997. Contrasting roles for the post-productivist countryside. In: Paul Cloke, J.L. (Ed.), *Contested Countryside Culture*. Routledge, London, pp. 70–93.
- Harman, B.P., Pruetz, R., Houston, P., 2015. Tradeable development rights to protect peri-urban areas: lessons from the United States and observations on Australian practice. *J. Environ. Plan. Manag.* 58 (2), 357–381. <https://doi.org/10.1080/09640568.2013.859130>.
- Hassink, J., Agricola, H., Thissen, J., 2016. Participation rate of farmers in different multifunctional activities in The Netherlands. *Outlook Agric.* 45 (3), 192–198. <https://doi.org/10.1177/0030727016665427>.
- Heasman, M., Lang, T., 2015. Food Wars: the Global Battle for Mouths, Minds and Markets. Routledge.
- Hellerstein, D., Nickerson, C., Cooper, J., Feather, P., Gadsby, D., et al., 2002. Farmland Protection: the Role of Public Preferences for Rural Amenities. US Department of Agriculture, Economic Research Service, Washington, DC.
- Hendrickson, J.R., Hanson, J.D., Tanaka, D.L., Sassenrath, G., 2008a. Principles of integrated agricultural systems: introduction to processes and definition. *Renew. Agric. Food Syst.* 23 (4), 265–271. <https://doi.org/10.1017/S1742170507001718>.
- Hendrickson, J.R., Liebig, M.A., Sassenrath, G.F., 2008. Environment and integrated agricultural systems. *Renew. Agric. Food Syst.* 23 (4), 304–313. <https://doi.org/10.1017/S1742170508002329>.
- Hodge, I., 2004. Methodology and action: economic rationales and agri-environmental policy choices. In: Brouwer, F. (Ed.), *Sustaining Agriculture and the Rural Environment*. Edward Elgar Publishing, Cheltenham, UK; Northampton, MA, USA.
- Hollier, C., Reid, M., 2007. *Small Farms: Valued Contributors to Healthy Rural Communities*. Rural Industries Research and Development Corporation.
- Holmes, J., 2006. Impulses towards a multifunctional transition in rural Australia: gaps in the research agenda. *J. Rural Stud.* 22 (2), 142–160. <https://doi.org/10.1016/j.jrurstud.2005.08.006>.
- Holmes, J., 2008. Impulses towards a multifunctional transition in rural Australia: interpreting regional dynamics in landscapes, lifestyles and livelihoods. *Landsc. Res.* 33 (2), 211–223. <https://doi.org/10.1080/01426390801912089>.
- Horlings, L.G., Marsden, T.K., 2011. Towards the real green revolution? Exploring the conceptual dimensions of a new ecological modernisation of agriculture that could 'feed the world'. *Glob. Environ. Chang.* 21 (2), 441–452. <https://doi.org/10.1016/j.gloenvcha.2011.01.004>.
- Houston, P., 2005. Re-valuing the fringe: some findings on the value of agricultural production in Australia's peri-urban regions. *Geogr. Res.* 43 (2), 209–223. <https://doi.org/10.1111/j.1745-5871.2005.00314.x>.
- Iaquinta, D.L., Drescher, A.W., 2000. Defining periurban: understanding rural-urban linkages and their connection to institutional contexts. In: Paper presented at the Tenth World Congress of the International Rural Sociology Association, Rio de Janeiro.

- Jackson, T., Zammit, K., Hatfield-Dodds, S., 2018. Snapshot of Australian Agriculture. [accessed November, 2018], from: <http://www.agriculture.gov.au/abares/Documents/snapshot-australian-agriculture.pdf>.
- Jongeneel, R.A., Polman, N.B.P., Slangen, L.H.G., 2008. Why are Dutch farmers going multifunctional. *Land Use Policy* 25, 81–94. <https://doi.org/10.1016/j.landusepol.2007.03.001>.
- Jurgilevich, A., Birge, T., Kentala-Lehtonen, J., Korhonen-Kurki, K., Pietikäinen, J., et al., 2016. Transition towards circular economy in the food system. *Sustainability* 8 (1), 69. <https://doi.org/10.3390/su8010069>.
- Kimbrell, A., 2002. *The Fatal Harvest Reader: the Tragedy of Industrial Agriculture*. Island Press, Washington, Covelo, London.
- Lawrence, G., Richards, C., Lyons, K., 2013. Food security in Australia in an era of neoliberalism, productivism and climate change. *J. Rural Stud.* 29, 30–39. <https://doi.org/10.1016/j.jrurstud.2011.12.005>.
- Lockie, S., 2009. Agricultural biodiversity and neoliberal regimes of agri-environmental governance in Australia. *Curr. Sociol.* (3), 407–426.
- Loff, B., Crammond, B., McConnell, C., 2010. Scoping a Food Policy Coalition. Victorian Health Promotion Foundation VicHealth. www.vichealth.vic.gov.au.
- Lovell, S.T., 2010. Multifunctional urban agriculture for sustainable land use planning in the United States. *Sustainability* 2 (8), 2499–2522. <https://doi.org/10.3390/su2082499>.
- Maier, L., Shobayashi, M., 2001. Multifunctionality - towards an Analytical Framework Agriculture and Food. Organisation for Economic Co-Operation and Development, Paris, France.
- Malcolm, L.R., 1996. *Agriculture in Australia: an Introduction*. Oxford University Press, Melbourne.
- Marsden, T.K., 2003. *The Condition of Rural Sustainability. European Perspectives on Rural Development*, Assen, Netherlands (Royal van Gorcum).
- Marsden, T.K., Sonnino, R., 2008. Rural development and the regional state: denying multifunctional agriculture in the UK. *J. Rural Stud.* 24 (4), 422–431. <https://doi.org/10.1016/j.jrurstud.2008.04.001>.
- Mathera, A.S., Hill, G., Nijnik, M., 2006. Post-productivism and rural land use: cul de sac or challenge for theorization? *J. Rural Stud.* 22, 441–455. <https://doi.org/10.1016/j.jrurstud.2006.01.004>.
- McCarthy, J., 2008. Rural geography: globalizing the countryside. *Prog. Hum. Geogr.* 32, 129–137. <https://doi.org/10.1177/0309132507082559>.
- McFarland, P., 2015. Multi-functionality and the urban-rural dichotomy in Australian metropolitan planning. In: Paper Presented at the State of Australian Cities Conference, Abstract retrieved from: <http://apo.org.au/system/files/63247/apo-nid63247-64861.pdf>.
- McKinna, D., Wall, C., 2014. *The Food and Beverage Growth Plan – Melbourne's North*. NORTH Link: Northern Melbourne Regional Development Australia (RDA) Committee.
- Meraner, M., Heijman, W., Kuhlman, T., Finger, R., 2015. Determinants of farm diversification in The Netherlands. *Land Use Policy* 42, 767–780. <https://doi.org/10.1016/j.landusepol.2014.10.013>.
- Methorst, R.G., Roep, D., Verhees, F.J.H.M., Verstegen, J.A.A.M., 2017. Differences in farmers' perception of opportunities for farm development. *NJAS - Wageningen J. Life Sci.* 81, 9–18. <https://doi.org/10.1016/j.njas.2017.02.001>.
- MMBW, 1971. *Planning Policies for Metropolitan Melbourne*. Melbourne and Metropolitan Board of Works.
- MMBW, 1981. *Metropolitan Strategy Implementation*. Melbourne and Metropolitan Board of Works, Melbourne, pp. 85–88.
- Morgan, S.L., Marsden, T.K., Miele, M., Morley, A., 2010. Agricultural multifunctionality and farmer's entrepreneurial skills: a study of Tuscan and Welsh farmers. *J. Rural Stud.* 26, 116–129. <https://doi.org/10.1016/j.jrurstud.2009.09.002>.
- OECD, 1979. *Agriculture in the Planning and Management of Peri-Urban Areas*. Organisation for Economic Co-operation and Development, Paris.
- Pagotto, M., Halog, A., 2016. Towards a circular economy in Australian agri-food industry: an application of input-output oriented approaches for analyzing resource efficiency and competitiveness potential. *J. Ind. Ecol.* 20, 1176–1186. <https://doi.org/10.1111/jiec.12373>.
- Paül, V., McKenzie, F.H., 2013. Peri-urban farmland conservation and development of alternative food networks: insights from a case-study area in metropolitan Barcelona (Catalonia, Spain). *Land Use Policy* 30, 94–105. <https://doi.org/10.1016/j.landusepol.2012.02.009>.
- Perrin, C., Nougarede, B., Sini, L., Brandiuni, P., Salvati, L., 2018. Governance changes in peri-urban farmland protection following decentralisation: a comparison between Montpellier (France) and Rome (Italy). *Land Use Policy* 70, 535–546. <https://doi.org/10.1016/j.landusepol.2017.09.027>.
- Petersen, B., Snapp, S., 2015. What is sustainable intensification? View from experts. *Land Use Policy* 46, 1–10. <https://doi.org/10.1016/j.landusepol.2015.02.002>.
- Pfeifer, C., Jongeneel, R.A., Sonneveld, M.P.W., Stoorvogel, J.J., 2009. Landscape properties as drivers for farm diversification: a Dutch case study. *Land Use Policy* 26, 1106–1115. <https://doi.org/10.1016/j.landusepol.2009.01.007>.
- Piotti, A., Zasada, I., Doernberg, A., Zoll, F., Rammé, W., 2018. Research for AGRI Committee—Urban and Peri-Urban Agriculture in the EU. Policy Department for Structural and Cohesion Policies, European Parliament, Brussels.
- Potter, C., Burney, J., 2002. Agricultural multifunctionality in the WTO: legitimate non-trade concern or disguised protectionism? *J. Agric. Econ.* 47, 172–190. [https://doi.org/10.1016/S0743-0167\(01\)00031-6](https://doi.org/10.1016/S0743-0167(01)00031-6).
- Pretty, J., 2008. Agricultural sustainability: concepts, principles and evidence. *Phil. Trans. R. Soc. B* 363, 447–465. <https://doi.org/10.1098/rstb.2007.2163>.
- Pritchard, B., McManus, P., 2000. *Land of Discontent*. UNSW Press, Sydney.
- Rawnsley, T., 2017. *Economic Performance of Australia's Cities and Regions*. SGS Economics & Planning Pty Ltd.
- Renting, H., Oostindie, H., Laurent, C., Brunori, G., Rossi, A., et al., 2005. Multifunctionality of Activities, Plurality of Identities and New Institutional Arrangements. FRA: Multagri Project Synthesis report (final report).
- Renting, H., Rossing, W.A.H., Groot, J.C.J., van der Ploeg, J.D., Laurent, C., et al., 2009. Exploring multifunctional agriculture. A review of conceptual approaches and prospects for an integrative transitional framework. *J. Environ. Manag.* 90 (2), S112–S123. <https://doi.org/10.1016/j.jenvman.2008.11.014>.
- Richardson, J., 2007. Beyond fairness: what really works to protect farmland. *Drake J. Agric. Law* 12 (1), 163–184.
- Rojo, M.S., Moratalla, A.Z., Alonso, N.M., Jimenez, V.H., 2014. Pathways towards the integration of peri-urban agrarian ecosystems into the spatial planning system. *Ecol. Process.* 3 (13). <https://doi.org/10.1186/s13717-014-0013-x>.
- Romero, J., Melo, C., 2015. Spanish Mediterranean Huertas: theory and reality in the planning and management of peri-urban agriculture and cultural landscapes. *WIT Trans. Ecol. Environ.* 193, 585–595. <https://doi.org/10.2495/SDP150501>.
- Rønningen, K., Flø, B., Fjeldavli, E., 2018. *The Legitimacy of a Multifunctional Agriculture*. Centre for Rural Research Norwegian University of Science and Technology.
- Sandercock, L., 1990. In: *Property, Politics, and Urban Planning: a History of Australian City Planning*, second ed. ed. New Brunswick (U.S.A.): Transaction Publishers, pp. 1890–1990 1990.
- Sassenrath, G.F., Heilman, P., Luschei, E., Bennett, G.L., Fitzgerald, G., et al., 2008. Technology, complexity and change in agricultural production systems. *Renew. Agric. Food Syst.* 23 (4), 285–295. <https://doi.org/10.1017/S174217050700213X>.
- Scherom, P., Soulard, C.T., 2018. The landscapes of professional farms in mid-sized cities, France. *Geogr. Res.* (2), 154. <https://doi.org/10.1111/1745-5871.12272>.
- Senate, 2016. *Scrutiny of Financial Advice Part I – Land Banking: a Ticking Time Bomb*. Economics References Committee. Commonwealth of Australia.
- Seunke, P., Lans, T., Wiskerke, J.S.C., 2013. Moving beyond entrepreneurial skills: key factors driving entrepreneurial learning in multifunctional agriculture. *J. Rural Stud.* 32, 208–219. <https://doi.org/10.1016/j.jrurstud.2013.06.001>.
- Simon, D., 2008. Urban environments: issues on the peri-urban fringe. *Annu. Rev. Environ. Resour.* 33, 167–185. <https://doi.org/10.1146/annurev.enviro.33.021407.093240>.
- Sokolow, A.D., 2004. *Optional Policies and Tools for Farmland and Open Space Protection in California*. Department of Human and Community Development, Davis University of California.
- Spataru, A., Faggiani, R., Sposito, V.A., Docking, A., 2018. Agricultural land suitability analysis of metropolitan peri-urban areas now and into the future — case study of city of Whittlesea, Melbourne, Australia. *Modern Environ. Sci. Eng.* 4 (10), 961–970. [https://doi.org/10.15341/mese\(2333-2581\)/10.04.2018/006](https://doi.org/10.15341/mese(2333-2581)/10.04.2018/006).
- Sturzak, J., Mell, I., 2017. *Green Belts: Past; Present; Future?* Routledge, New York, London.
- Tilzey, M., 2006. Neo-liberalism, the WTO and new modes of agri-environmental governance in the European Union, the USA and Australia. *Int. J. Sociol. Agric. Food* 14 (1), 1–28.
- Tonts, M., Grieve, S., 2002. Commodification and creative destruction in the Australian rural landscape: the case of Bridgetown, western Australia. *Aust. Geogr. Stud.* 40 (1), 58–70. <https://doi.org/10.1111/1467-8470.00161>.
- Tsutsumi, J., Wyatt, R., 2006. A brief history of metropolitan planning in Melbourne, Australia. *Appl. GIS* 2 (2), 1–10.
- van der Ploeg, J.D., Long, A., Banks, J., 2002. *Living Countrysides. Rural Development Processes in Europe: the State of the Art*. Elsevier, Detinchem.
- Van Huylenbroeck, G., Durand, G., 2003. *Perspectives on Rural Policy and Planning Multifunctional Agriculture. A New Paradigm for European Agriculture and Rural Development*. Ashgate, Aldershot.
- Van Huylenbroeck, G., Vandermeulen, V., Mettepenningen, E., Verspecht, A., 2007. Multifunctionality of agriculture: a review of definitions, evidence and instruments. *Living Rev. Landscape Res.* 1, 3. <https://doi.org/10.12942/lrlr-2007-3>.
- Vandermeer, J., Lawrence, D., Symstad, A., Hobbie, S., 2002. Effects of biodiversity on ecosystem functioning in managed ecosystems. In: Loreau, M., Naeem, S., Inchausti, P. (Eds.), *Biodiversity and Ecosystem Functioning*. Oxford University Press, Oxford, UK, pp. 157–168.
- Vinge, H., 2018. Farmland conversion to fight climate change? Resource hierarchies, discursive power and ulterior motives in land use politics. *J. Rural Stud.* 64, 20–27. <https://doi.org/10.1016/j.jrurstud.2018.10.002>.
- VPP, 2018. *Victoria Planning Provisions*. [accessed February, 2019], from: http://planning-schemes.delwp.vic.gov.au/_data/assets/pdf_file/0007/481723/VPPs_All_Clauses.pdf.
- Wilson, G.A., 2007. *Multifunctional Agriculture: a Transition Theory Perspective*. CABI, Wallingford, UK.
- Wilson, G.A., 2008. From 'weak' to 'strong' multifunctionality: conceptualising farm-level multifunctional transitional pathways. *J. Rural Stud.* 24, 367–383. <https://doi.org/10.1016/j.jrurstud.2007.12.010>.
- Wu, T.-c., 2018. Agriculture tourism and the transformation of rural countryside. *Tour. Geogr.* 20 (2), 354–357. <https://doi.org/10.1080/14616688.2018.1434819>.